# INTRODUCTION TO PROGRAMMING

| Course Code | Course Title | L | T | P | S | J | C |
|---|---|---|---|---|---|---|---|
| | | 1 | 0 | 2 | 0 | 0 | 2 |
| Pre-requisite | NIL | | | | | | |
| Co-requisite | NIL | | | | | | |
| Preferable Exposure | NIL | | | | | | |

## Course Description

Introduction to Programming is a foundational course designed to help undergraduate students develop basic problem-solving and programming skills. The course introduces the fundamentals of computer science, algorithmic thinking, and flowchart-based problem-solving using RAPTOR. It also provides an introduction to Python programming, covering syntax, variables, operators, control structures, data structures, and functions. Through practical programming exercises, students will learn to design algorithms, write Python programs, and apply programming concepts to solve simple computational problems effectively.

## Course Objectives

- To understand the basic concepts of computer science and programming.
- To develop problem-solving skills using algorithms and flowcharts.
- To learn the fundamentals of Python syntax, variables, and operators.
- To apply control structures for solving simple programming problems.
- To use data structures and functions in Python programs.

## MODULE -I

**9** Hours

**1.** Introduction to Programming and Computer Science - Overview of computer science, Role of programming in computer science, Types of programming languages, Problem-solving techniques

**FLOW CHARTS**

2. Flow Charts and Problem-Solving using Raptor, Introduction to the idea of an algorithm and Flow charts, Flow chart symbols, Input/Output, Assignment, operators.

Example problems on Flow Charts, Finding Maximum and Minimum of n numbers, Unit converters, Interest calculators, multiplication tables, GCD of 2 numbers, Fibonacci number generation, prime number generation, Prime Number checking , average of n numbers

## MODULE -II

**BASICS OF PYTHON PROGRAMMING** **9** Hours

Introduction to Python: Features of Python, Why Python for programming?

Writing your first Python program

Basic Syntax and Variables: Keywords and Identifiers, Variables and Types (int, float, string), Basic Operators and Expressions and its program examples

## MODULE -III

**CONTROL STRUCTURES** 9 Hours

Conditional Statements: if, if-else, if-elif-else and its program examples

Simple loops: while, for, for -each, and its program examples

## MODULE -IV

**DATA STRUCTURES** 9 Hours

Lists, Tuples, Sets, Dictionaries Operations on each data structures and its program examples

## MODULE -5

9 Hours

Functions: Defining and calling functions, Parameters and return values, Scope and lifetime of variables.

**List of Programs**

Program to demonstrate data types

Program to demonstrate expressions

Program to demonstrate Arithmetic Operators, Assignment Operators, Comparison Operators with input statements

Program to demonstrate Logical Operators, Bitwise Operators, Membership Operators, Assignment Operators, Comparison Operators with input statements .

Program to sum a power series upto a specified significant figure examples Geometric Series ,generate sin(x),cos(x),e power x series.

Program to demonstrate various operations on Lists

Program to demonstrate various operations on Tuples

Program to demonstrate various operations on Sets

Program to demonstrate various operations on Dictionaries

Program to demonstrate Bubble Sort

Program : Basic Function Definition and Call

Write a program to create a function named greet that takes a name as a parameter and prints a greeting message.

Program : Function with Return Value

Write a functions called Sum/Difference /Product/ Quotient that takes two numbers as parameters and returns their Sum/Difference /Product/ Quotient

Program : Multiple Return Values

Write a program to create a function named stats that takes a list of numbers and returns the minimum, maximum, and average of the list.

Program : Variable Scope

Write a program to demonstrate the difference between local and global variables in a function.

Program: Lifetime of Variables

Write a program to show how the lifetime of a local variable is limited to the function's execution.

**Textbooks**

1. A.B. Chaudhuri, *Flowchart and Algorithm Basics: The Art of Programming*, Mercury Learning and Information, 2020.
2. Charles Dierbach, *Introduction to Computer Science Using Python: A Computational Problem-Solving Focus*, Wiley.
3. John M. Zelle, *Python Programming: An Introduction to Computer Science*, 4th Edition, Shroff/Franklin, Beedle, 2024.

**Suggested Reference Books**

1. Eric Matthes, *Python Crash Course*, 3rd Edition, No Starch Press, 2023.
2. Allen B. Downey, *Think Python* ,Green Tea Press.
3. Richard E. Pattis, *Flowcharting & Algorithmic Thinking*, 2nd Edition

**Course Outcomes**

At the end of the course, the student is able to:

**CO1:** Understand basic concepts of programming, algorithms, and flowcharts.

**CO2:** Develop Python programs using syntax, variables, data types, and operators.

**CO3:** Apply conditional statements and loops to solve problems.

**CO4:** Use Python data structures for storing and manipulating data.

**CO5:** Write modular programs using functions and variable scope.

CO-PO-PSO Mapping Table

| | POs | PSOs |
|---|---|---|

| CO | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 2 | 1 | 1 | 2 | - | - | - | 2 | 1 | - | - | - | - | - | - | 2 | 1 | - | - |
| 2 | 3 | 2 | 2 | 1 | 3 | - | - | - | 2 | 3 | - | - | 1 | - | - | - | 3 | 2 | 1 | - |
| 3 | 3 | 3 | 2 | 1 | 3 | 1 | - | - | 2 | 2 | - | - | 1 | - | - | - | 3 | 3 | 1 | - |
| 4 | 3 | 2 | 2 | 1 | 3 | 1 | - | - | 2 | 3 | - | - | 1 | - | - | - | 3 | 3 | 1 | - |
| 5 | 3 | 3 | 2 | 2 | 3 | 2 | 2 | 1 | 3 | 3 | - | 1 | 2 | - | - | - | 3 | 3 | 2 | 1 |

3 – High, 2 – Medium & 1 – Low Correlation

## SDG No(s). & Statement(s) :

| | |
|---|---|
| **APPROVED IN:** | |
| **BOS : 21.4.2026** | **ACADEMIC COUNCIL:30.4.2026** |
| | |
| **SDG No. & Statement:** | |
| 4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all. | |
| | |
| **SDG Justification:** | |
| The topics included in this course are designed to get acquainted with one of the skills that handle necessary mathematical orientation, programming techniques and concept based learning. | |